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HR Data Analysis

Employee Attrition and Retention Analysis
(Workforce Insights)

SQL + Power BI | Data-Driven HR Analytics

Problem Statement

Organizations often face challenges with high employee attrition, leading to increased recruitment costs, reduced productivity, and workforce instability. This project aims to help HR departments analyze employee data using SQL, Power BI, and Power Query to identify the key factors influencing employee turnover, such as salary, job role, experience, education field, and demographics. Through interactive dashboards and analytical insights, the solution supports data-driven HR decision-making for improving employee retention, workforce planning, and overall organizational performance.

Business Impact (HR-Focused & Actionable)

- Which employee groups are most likely to leave?
- Which factors combined drive early attrition?
- Where should retention budgets be prioritized?
- How can we identify at-risk employees before they resign?
- What role do promotion cycles play in turnover?

Attrition rate by Gender

```sql
-- Attrition rate by Gender
SELECT
    Gender,
    COUNT(*) AS Total_Employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrition_Count,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate_Pct
FROM hr_analytics
GROUP BY Gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Gender	Total_Employees	Attrition_Count	Attrition_Rate_Pct
Male	849	145	17.08
Female	567	84	14.81

Attrition by Over Time

```sql
-- Attrition by Over Time
select
OverTime,
    COUNT(*) AS Total,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
FROM hr_analytics
GROUP BY OverTime;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

OverTime	Total	Attrited	Attrition_Rate
No	1011	104	10.29
Yes	405	125	30.86

Department-wise attrition rate with ranking

```sql
-- Department-wise attrition rate with ranking

SELECT
    Department,
    COUNT(*) AS Total,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate,
    RANK() OVER (ORDER BY ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) DESC) AS Risk_Rank
FROM hr_analytics
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	Total	Attrited	Attrition_Rate	Risk_Rank
Sales	434	90	20.74	1
Human Resources	62	12	19.35	2
Research & Development	920	127	13.80	3

Top 5 job role by highest attrition count

```sql
-- Top 5 job role by highest attrition count
select
JobRole,
    COUNT(*) AS Attrition_Count
FROM hr_analytics
WHERE Attrition = 'Yes'
GROUP BY JobRole
ORDER BY Attrition_Count DESC
LIMIT 5;
```

Result Grid   Filter Rows: Export: Wra

JobRole	Attrition_Count
Laboratory Technician	60
Sales Executive	55
Research Scientist	44
Sales Representative	33
Human Resources	12

Attrition rate by job satisfaction level (1-4)

```sql
-- Attrition rate by job satisfaction level (1-4)

select
JobSatisfaction,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
FROM hr_analytics
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

JobSatisfaction	Attrition_Rate
1	22.55
2	16.79
3	16.63
4	11.36

Job role + Overtime combination with highest attrition

```sql
-- Job role + Overtime combination with highest attrition
SELECT
    JobRole,
    OverTime,
    COUNT(*) AS Attrition_Count
FROM hr_analytics
WHERE Attrition = 'Yes'
GROUP BY JobRole, OverTime
ORDER BY Attrition_Count DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

JobRole	OverTime	Attrition_Count
Research Scientist	Yes	31
Laboratory Technician	Yes	31
Sales Executive	Yes	31
Laboratory Technician	No	29
Sales Executive	No	24
Sales Representative	No	17
Sales Representative	Yes	16
Research Scientist	No	13
Human Resources	No	7
Healthcare Representative	No	7

Average years at company for attrited vs active

```sql
-- Average years at company for attrited vs active

SELECT
    Attrition,
    ROUND(AVG(YearsAtCompany), 2) AS Avg_Years_At_Company
FROM hr_analytics
GROUP BY Attrition;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Attrition	Avg_Years_At_Company
Yes	5.10
No	7.41

Attrition by Enviorment Satisfaction

```sql
-- Attrition by Enviorment Satisfaction

SELECT
    EnvironmentSatisfaction,
    COUNT(*) AS Total,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
FROM hr_analytics
GROUP BY EnvironmentSatisfaction
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EnvironmentSatisfaction	Total	Attrited	Attrition_Rate
1	270	69	25.56
2	278	40	14.39
3	438	62	14.16
4	430	58	13.49

Attrition by Years Since Last Promotion

```sql
-- Attrition by Years Since Last Promotion

SELECT
    YearsSinceLastPromotion,
    COUNT(*) AS Attrited_Count
FROM hr_analytics
WHERE Attrition = 'Yes'
GROUP BY YearsSinceLastPromotion
ORDER BY Attrited_Count DESC
LIMIT 5;
```

esult Grid | Filter Rows: | Export: | W

YearsSinceLastPromotion	Attrited_Count
0	106
1	47
2	27
7	16
3	9

High Performer Attrition Risk

```sql
-- High Performer Attrition Risk

SELECT
    `EmpID`,`JobRole`,Department,`PerformanceRating`,`MonthlyIncome`,`YearsAtCompany`,
    (
        CASE WHEN `PerformanceRating` >= 4 THEN 3 ELSE 0 END +
        CASE WHEN `JobSatisfaction` <= 2 THEN 3 ELSE 0 END +
        CASE WHEN `PercentSalaryHike` < 13 THEN 2 ELSE 0 END +
        CASE WHEN `OverTime` = 'Yes' THEN 2 ELSE 0 END +
        CASE WHEN `YearsSinceLastPromotion` >= 3 THEN 1 ELSE 0 END
    ) AS High_Performer_Exit_Risk
FROM hr_analytics
WHERE Attrition = 'No'
ORDER BY High_Performer_Exit_Risk desc
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

EmpID	JobRole	Department	PerformanceRating	MonthlyIncome	YearsAtCompany	High_Performer_Exit_Risk
RM452	Manufacturing Director	Research & Development	4	7406	10	9
RM919	Manager	Sales	4	19847	29	9
RM883	Manufacturing Director	Research & Development	4	10252	7	9
RM278	Sales Executive	Sales	4	5605	8	9
RM887	Research Scientist	Research & Development	4	3579	11	9
RM613	Sales Executive	Sales	4	4779	8	9
RM264	Manager	Sales	3	16872	7	8
RM579	Manufacturing Director	Research & Development	3	5980	15	8
RM928	Manufacturing Director	Research & Development	3	5410	16	8
RM1316	Research Scientist	Research & Development	4	6962	1	8

Attrition trend by years at company (detailed)


```sql
-- Attrition trend by years at company (detailed)

select
    YearsAtCompany,
    COUNT(*) AS Total_Employees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attrited,
    ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS Attrition_Rate
FROM hr_analytics
GROUP BY YearsAtCompany
ORDER BY YearsAtCompany asc
limit 10;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

YearsAtCompany	Total_Employees	Attrited	Attrition_Rate
0	42	16	38.10
1	162	57	35.19
2	123	26	21.14
3	119	20	16.81
4	106	18	16.98
5	192	20	10.42
6	74	9	12.16
7	90	11	12.22
8	78	9	11.54
9	79	8	10.13

Conclusion

The Employee Attrition and Retention Analysis project was developed to help HR teams identify the key reasons behind employee turnover and improve workforce retention. Using SQL, Power BI, and Power Query, workforce data was analyzed through business-driven SQL queries and interactive dashboards to uncover insights related to overtime, job role, experience, and employee satisfaction. The project supports data-driven HR decision-making by helping organizations reduce attrition, improve employee satisfaction, and optimize workforce planning.

Key HR Takeaways for Retention Strategy

· Sales & HR are crisis zones: They have the highest attrition rates (20.74% and 19.35%).

· Low satisfaction is a dealbreaker: Attrition rate jumps to 22.55% when job satisfaction is level 1.

· Pay is a baseline, not a solution: Most attrition happens in the lowest salary band.

· Watch your high-performers: Top talent is leaving due to low raises and lack of promotion.

· Overtime burns out critical roles: Especially high impact for Lab Technicians and Sales Executives.

· Stagnation predicts exit: If no promotion in 3+ years, attrition risk spikes.

· Target retention spend: Focus budget only on high performers, not all attrition.

· Flag men in specific roles: Male attrition count is higher and driven by specific jobs (Lab Tech, Sales).

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Thank You

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